

APPENDIX *B*

A Regularization Procedure

The following algorithm chooses parameters δ, γ that guarantee that the regularized primal-dual matrix (19.25) is nonsingular and satisfies the inertia condition (19.24). The algorithm assumes that, at the beginning of the interior-point iteration, δ_{old} has been initialized to zero.

Algorithm B.1 (Inertia Correction and Regularization).

Given the current barrier parameter μ , constants $\eta > 0$ and $\beta < 1$, and the perturbation δ_{old} used in the previous interior-point iteration.

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Factor (19.25) with  $\delta = \gamma = 0$ .
if (19.25) is nonsingular and its inertia is  $(n + m, l + m, 0)$ 
    compute the primal-dual step; stop;
if (19.25) has zero eigenvalues
    set  $\gamma \leftarrow 10^{-8} \eta \mu^\beta$ ;
if  $\delta_{old} = 0$ 
    set  $\delta \leftarrow 10^{-4}$ ;

else
    set  $\delta \leftarrow \delta_{old}/2$ ;
repeat
    Factor the modified matrix (19.25);
    if the inertia is  $(n + m, l + m, 0)$ 
        Set  $\delta_{old} \leftarrow \delta$ ;
        Compute the primal-dual step (19.12) using the coefficient
            matrix (19.25); stop;
    else
        Set  $\delta \leftarrow 10\delta$ ;
end (repeat)

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This algorithm has been adapted from a more elaborate procedure described by Wächter and Biegler [301]. All constants used in the algorithm are arbitrary; we have provided typical choices. The algorithm aims to avoid unnecessarily large modifications δI of $\nabla_{xx}^2 \mathcal{L}$ while trying to minimize the number of matrix factorizations. Excessive modifications degrade the performance of the algorithm because they erase the second derivative information contained in $\nabla_{xx}^2 \mathcal{L}$, and cause the step to take on steepest-descent like characteristics. The first trial value ($\delta = \delta_{old}/2$) is based on the previous modification δ_{old} because the minimum perturbation δ required to achieve the desired inertia will often not vary much from one interior-point iteration to the next.

The heuristics implemented in Algorithm B.1 provide an alternative to those employed in Algorithm 7.3, which were presented in the context of unconstrained optimization. We emphasize, however, that all of these are indeed heuristics and may not always provide adequate safeguards.